John Henry Mejia

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Education

Texas Christian University August 2020 - May 2024
B.S. in Computer Science Trustee Scholar and Chancellor's Scholar
Minors: Mathematics, General Business UPE Computing Honor Society

Experience

Lockheed Martin Co - F-22 IMIS - Systems Engineer May 2024 - Present

- Collaborated across software teams to create high-level and component-level diagrams in Cameo, develop system requirements, and support process improvements using Agile.
- Modeled system releases in Cameo by refactoring and creating diagrams (activity, sequence, parametric, use case, and block definition) to ensure accurate system representation.
- Authored CONOPS, System Design, Interface Design, and Software Design Documents. Wrote and executed application- and system-level procedures for verification and validation (V&V). Assisted in formal Preliminary Design Reviews (PDR).
- Led the requirements lifecycle as Requirements Manager. Presented changes to Requirements Review Board and drove development, maintenance, and retirement of requirements.
- Administered, connected, and standardized DOORS and Cameo databases across unclassified and classified/air-gapped systems, maintaining data integrity. Used DXL/SQL/Java/Python to streamline document exports by 94% and enable document generation from the Cameo model.
- VEGA Stretch Project: Built backstage.io portal for developers on closed network. Enabled self-service provisioning of VM instances. Implemented plugins for trend reports on GitLab pipelines/re-run pipelines, RBAC, Azure cost data, Grafana dashboards, JIRA trends, and DORA metrics. Developed plugins to edit and display custom program information.

Atlassian - Site Reliability Engineer Intern May 2023 - Aug 2023

- Migrated runbook backup system to AWS SQS + Lambdas, improving scalability, reliability and decreasing backup time by ∼90.2%, allowing the service to be run 4x as often and decreasing cost by 71.4% week-over-week.
- Backup runbook system incrementally backs up 14,700 runbooks daily, supporting 188 separate spaces (teams).
- Streamlined database health monitoring through automated generation of PDF reports using AWS S3, Lambdas, and Step Functions; reports sent to Confluence and unhealthy signs are alerted via Slack.

Ultra Electronics - Software Engineer Intern May 2022 - Aug 2022

- Developed plane clustering algorithms and flight pattern identifiers to enhance combat ID recognition in the RAIN (Rapid Application of Information) system. (sklearn DBSCAN, OpenCV feature extraction)
- Maintained Jenkins pipelines to streamline Grype scans of Linux container images, automated response emails.
- Created Java application to transform between different coordinate systems from different flight data sources.

Projects and Research

Team Leader, AI-Enhanced Educational Portal (Senior Design) 2023-2024 · https://classifai.tcu.edu/

- Managed a team to automate transcription and question analysis of classroom audio for data-driven feedback.
- Created automated transcription/diarization system through OpenAI Whisper and Pyannote. Categorized questions according to Costa's levels of thinking via Meta's LLaMA-3.2.
- Improved base accuracy of LLaMA question categorization by 8.7% (71% → 77.1%) through ORPO fine-tuning. All analysis done locally for cost-effectiveness and privacy.

Computer Skills

Languages & Databases: Python, UML, SysML 2, C#, C, C++, Java, DXL (DOORS), R, SQL, Bash
Frameworks/Tools: Flask, TensorFlow (Keras), Apache Velocity, Spring Boot, Jenkins, Kubernetes, Docker, VMWare, Cameo Systems Modeler, DOORS
Libraries/Cloud: NumPy, Pandas, matplotlib, scikit-learn, git, AWS SQS + Lambdas, AWS GovCloud, Azure DevOps, Linux/Unix

Other

TCU Rowing Team 2020 - Present: Coach; Former President; National Record Holder
Languages English (Fluent), Spanish (Fluent)
Security Clearance Secret